

“Transhumant Pastoralism, Climate Change and Conflict in Africa”

Eoin McGuirk and Nathan Nunn, Review of Economic Studies, 2024

README file for replication materials

A. Contents

This replication folder contains four subfolders:

1. analysis

This folder contains the Stata do-file (*EMG_NN_THPCCC_replication_code.do*) that reproduces the output presented in the manuscript and appendix.

2. data

This folder contains the Stata data file (*EMG_NN_THPCCC_replication_data.dta*) that is used by the replication do-file.

3. output

This folder contains the following:

- All tables in *.tex format.
- Stata log output in *.txt format (which is produced after running the replication code).
- *log_2023_07_10.txt*, the Stata log output that was produced by the replication code on July 11th, 2023.

4. paper

This folder contains a populated *.tex file that will produce a PDF of all tables after running the replication code. This folder also contains auxiliary latex files for formatting.

B. Instructions

In the do-file, insert relevant file path name after `global folder = ...`. This is the only change required for users to replicate all tables in the paper.

Running the file from top to bottom will produce (i) all tables in *.tex format and (ii) Stata output log in *.txt format in the output folder. Alternatively, users may search for (or scroll down to) code for individual tables.

Additional Notes

The replication code first reproduces the descriptive statistics tables, i.e., Tables A3, A4, and A16. Variable labels are initially assigned with a view to replicating these tables. Thereafter, tables are produced in the order in which they appear in the paper. Variable labels are updated where necessary table by table.

Note: labels do not explicitly indicate at what level each measure varies. In the tables, this is instead indicated by panel headings for Nearest Neighboring Ethnic Group, Own Ethnic Group, and Own Cell. However, where relevant, it is possible to determine the level of aggregation from the variable name in the data file. The prefix `n1_` indicates nearest neighboring ethnic group and `g_` indicates ethnic group. Thus, for example, `prec_gpcc` is cell (and year) level rain, `g_prec_gpcc` is ethnic group (and year) level rain, and `n1_g_prec_gpcc` is nearest neighboring ethnic group (and year) level rain. For variables that naturally vary at the level of an ethnic group (i.e., variables from the Ethnographic Atlas, such as the narrow measure of transhumant pastoralism `herdXEApxn12` or a group's dependence on agriculture `EA_ag`), the prefix `g_` is unnecessary.

C. Data Availability Statement and Data Citations

Data sources for all variables are described in detail in Appendix Section A of the manuscript. This includes citations and URLs for download where applicable. Unless otherwise stated below, datasets used in the analysis do not require payment or other significant qualifications (as of June 2023).

Below are the sources of data in order of presentation with citations (as suggested by sources, where applicable):

UCDP: Conflict

Sundberg, Ralph and Erik Melander, "Introducing the UCDP Georeferenced Event Dataset," *Journal of Peace Research*, 2013, 50 (4), 523–532.

URL as of Jan 2024: https://ucdp.uu.se/downloads/index.html#ged_global

ACLED: Conflict

Raleigh, Clionadh, Andrew Linke, Håvard Hegre, and Joakim Karlsen, "Introducing ACLED-Armed Conflict Location and Event Data," *Journal of Peace Research*, 2010, 47 (5), 651–660.

URL: <https://acleddata.com/data-export-tool/> [must generate free account]

Ethnographic Atlas and Murdock Map

Murdock, George Peter, *Africa: Its Peoples and Their Cultural History*, New York: McGraw-Hill Book Company, 1959.

___, *Ethnographic Atlas*, Pittsburgh: University of Pittsburgh Press, 1967.

GPCC: Rain

Schneider, Udo, Andreas Becker, Peter Finger, Elke Rustemeier, and Markus Ziese, "GPCC Full Data Monthly Product Version 2020 at 0.5°: Monthly Land-Surface Precipitation from Rain-Gauges built on GTS-based and Historical Data," 2020. Dataset.

URL: https://opendata.dwd.de/climate_environment/GPCC/html/fulldata-monthly_v2020_doi_download.html

Copernicus: Phytomass

Copernicus, "Copernicus Global Land Service: Dry Matter Productivity, Global, 1km, 1999-2019," n.d.

URL: <https://land.copernicus.eu/global/products/dmp>. Accessed Jan 22, 2021.

PRIO-GRID: Temperature and Population

Tollefsen, Andreas Forø, Håvard Strand, and Halvard Buhaug, "PRIO-GRID: A Unified Spatial Data Structure," *Journal of Peace Research*, 2012, 49 (2), 363–374.

URL: <https://grid.prio.org/#/download>

World Religion Database

Johnson, Todd M. and Brian J. Grim, "World Religion Database," 2021.

URL: <https://www.worldreligiondatabase.org/> [requires membership fee]

AidData: Foreign Aid

Tierney, Michael J., Daniel L. Nielson, Darren G. Hawkins, J. Timmons Roberts, Michael G. Findley, Ryan M. Powers, Bradley Parks, Sven E. Wilson, and Robert L. Hicks, "More Dollars than Sense: Refining Our Knowledge of Development Finance Using AidData," *World Development*, 2011, 39 (11), 1891–1906.

AidData, "AidDataCore_ResearchRelease_Level1_v3.1 Research Releases dataset," 2017. URL: <http://aiddata.org/research-datasets>. Accessed April 9, 2021.

Protected Planet: Conservation

UNEP-WCMC and IUCN, "Protected Planet: The World Database on Protected Areas (WDPA) and The World Database on Other Effective Area-based Conservation Measures (WD-OECM) [Online]," 2021. Available at: www.protectedplanet.net. Accessed May 14 2021.

Political Power: Ethnic Power Relations

Cederman, Lars-Erik, Andreas Wimmer, and Brian Min, "Why Do Ethnic Groups Rebel? New Data and Analysis," *World Politics*, 2010, 62 (1), 87–119.

URL: <https://icr.ethz.ch/data/epr/core/>

Segmentary Lineage

Moscona, Jacob, Nathan Nunn, and James A. Robinson, "Segmentary Lineage Organization and Conflict in Sub-Saharan Africa," *Econometrica*, 2020, 88 (5), 1999–2036.

URL:

<https://www.econometricsociety.org/publications/econometrica/browse/2020/09/01/segmentary-lineage-organization-and-conflict-sub-saharan-africa>

World Bank: Commodity Prices

World Bank, "World Bank Commodity Price Data (The Pink Sheet)," 2021. URL: <https://www.worldbank.org/en/research/commodity-markets>. Accessed Mar 18 2021.

MIRCA2000: Seasons

Portmann, Felix T., Stefan Siebert, and Petra Döll, "MIRCA2000—Global Monthly Irrigated and Rainfed Crop Areas Around the Year 2000: A New High-Resolution Data Set for Agricultural and Hydrological Modeling," *Global Biogeochemical Cycles*, 2010, 24 (1), GB1011.

URL: https://www.uni-frankfurt.de/45218031/Data_download_center_for_MIRCA2000

D. Computational and Software Requirements

The code was last run using StataMP 17 on a 4-core Intel-based laptop with MacOS version 11.6. Run time for the full do-file was approximately 4 hours 25min (see log_2023_07_10.txt.)

Code for installing additional software packages (estout, reghdfe, acreg) is included in EMG_NN_THPCC_replication_code.do